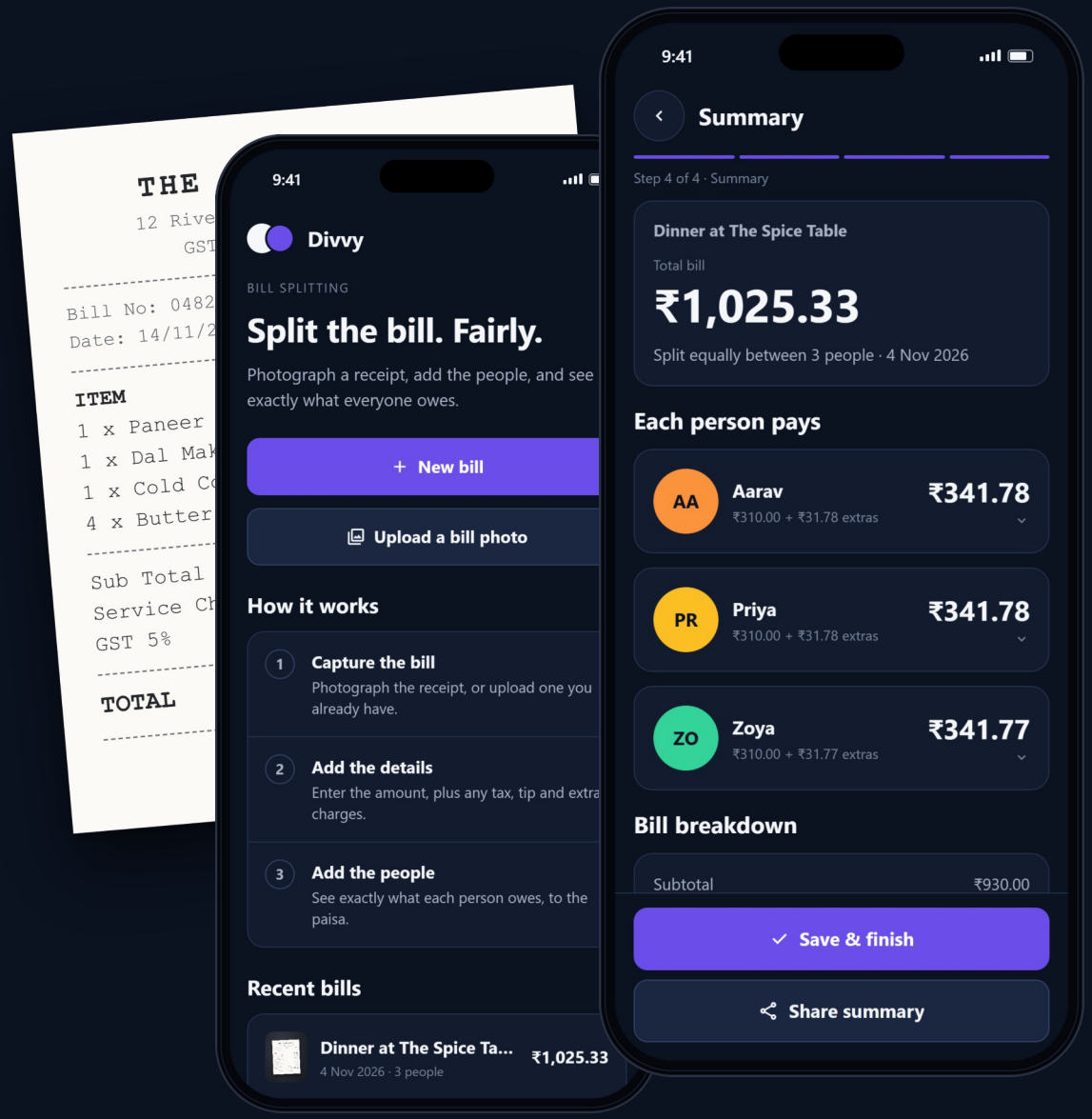
Divvy

MOBILE APPLICATION DEVELOPMENT · PROJECT

Split the bill. Fairly.

Divvy — intelligent, receipt-based bill splitting

Photograph a receipt, add the people, and see exactly what everyone owes — to the paisa.



Krish Dwivedi
BSc Computer Science · UIT, Karnavati University
Mobile Application Development · 2026

Meet Divvy

Divvy turns a photographed restaurant bill into a fair, exact split for everyone at the table.

What it is

A receipt-first mobile app for iPhone and Android, built with React Native and Expo.

The problem it solves

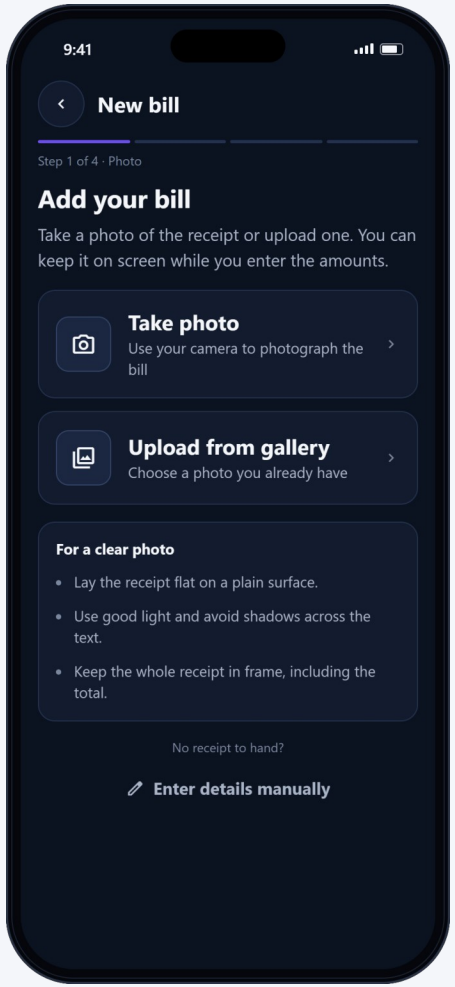
Shared bills are split by hand: slow, error-prone and hard to verify.

Who it is for

Friends, flatmates, students and colleagues who eat out and share costs.

Why it is useful

Every share is exact, explained line by line, and computed on the device.



APP SCREENSHOT · ADD YOUR BILL

Splitting a bill by hand is slow, unclear and often wrong

Manual calculations
Totals worked out on a calculator, at the table.

Allocating items
Equal splits ignore who actually ordered what.

Taxes and charges
Is GST charged on the service charge? Usually guessed.

Inaccurate shares
Rounding each share separately creates or loses paise.

No record
Once the chat moves on, nobody remembers who paid what.

BY HAND

₹1,025.33 ÷ 3

= 341.7766… → ₹341.78 each

AA	Aarav	₹341.78
PR	Priya	₹341.78
ZO	Zoya	₹341.78

Collected **₹1,025.34**

₹0.01 over the bill, and Zoya pays ₹14.71 for paneer she never ate.

WITH DIVVY

Item by item

Plus fair service charge and GST

AA	Aarav	₹349.15
PR	Priya	₹349.12
ZO	Zoya	₹327.06

Collected **₹1,025.33**

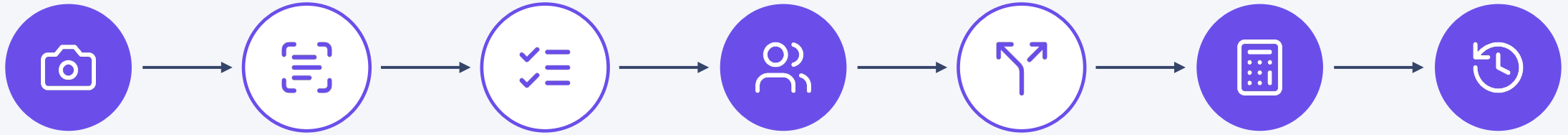
Exactly the bill. Each person pays for what they ordered.

Same sample bill: 4 items, service charge 5%, GST 5% — 3 people.

Implemented Final release

A receipt-first workflow, from photo to fair share

Divvy starts from the receipt itself. Each step hands clean, structured data to the next — and every value stays editable.



01
Capture
Photograph the bill or pick it from the gallery

02
Scan
On-device OCR reads items, prices and charges

03
Review
Check and correct every extracted value

04
People
Add everyone sharing the bill

05
Allocate
Assign each item to whoever ordered it

06
Calculate
Exact shares, including tax, tip and charges

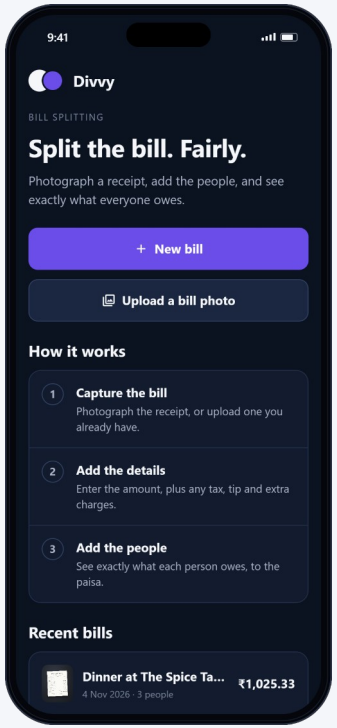
07
History
Save, revisit and share past bills

Receipt-first
The photo is the starting point of the bill, not an attachment.

Exact to the paisa
Shares always add up to the bill total — by construction.

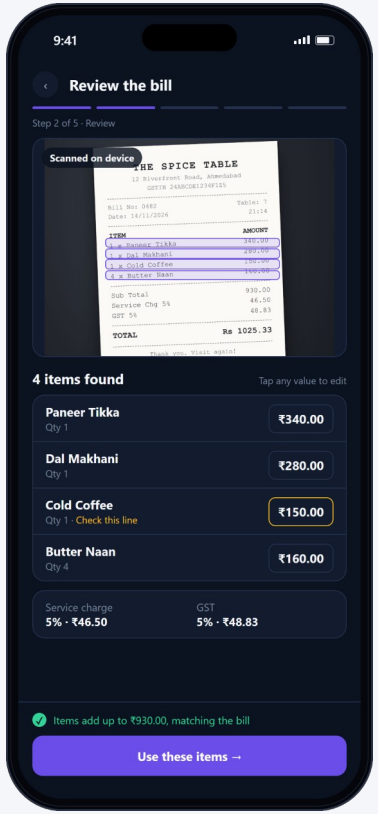
Private by design
Runs on the device. No account, nothing uploaded.

The Divvy app



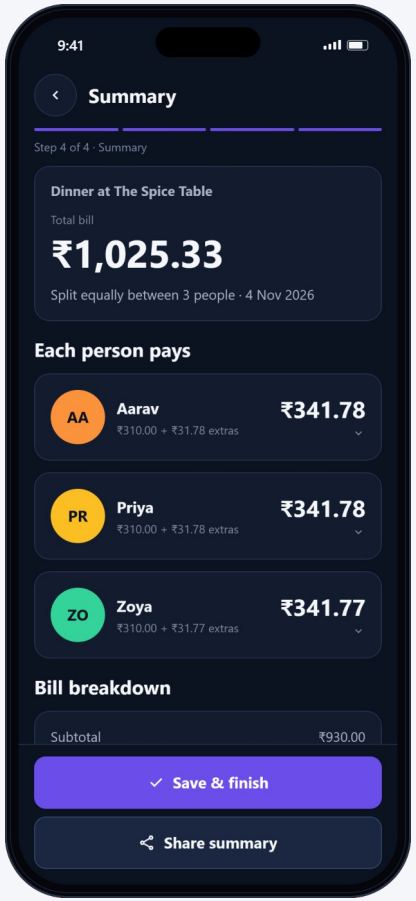
Home dashboard

APP SCREENSHOT



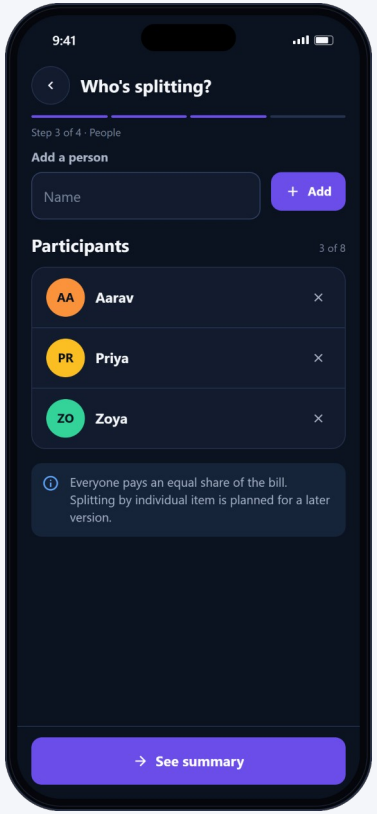
Receipt scanning

CONCEPT DESIGN



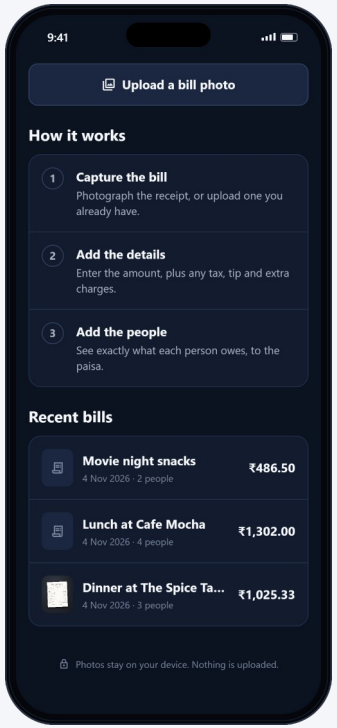
Bill summary

APP SCREENSHOT



Participants

APP SCREENSHOT



Bill history

APP SCREENSHOT

Implemented Final release

Everything needed to split a bill properly

Camera capture
Photograph a bill with the system camera

Gallery upload
Use a receipt photo you already have

Automatic extraction
On-device OCR reads items, prices and totals

Editable bill details
Amount, tax, tip and extra charges

Participants
Up to 8 people, each with a colour avatar

Item-wise allocation
Assign every item to whoever ordered it

Automatic splitting
Exact shares, recalculated instantly

Tax, tip and charges
Percent or fixed amount, applied in order

Individual breakdowns
See exactly how each share is made up

Bill history
Recent bills on the home screen

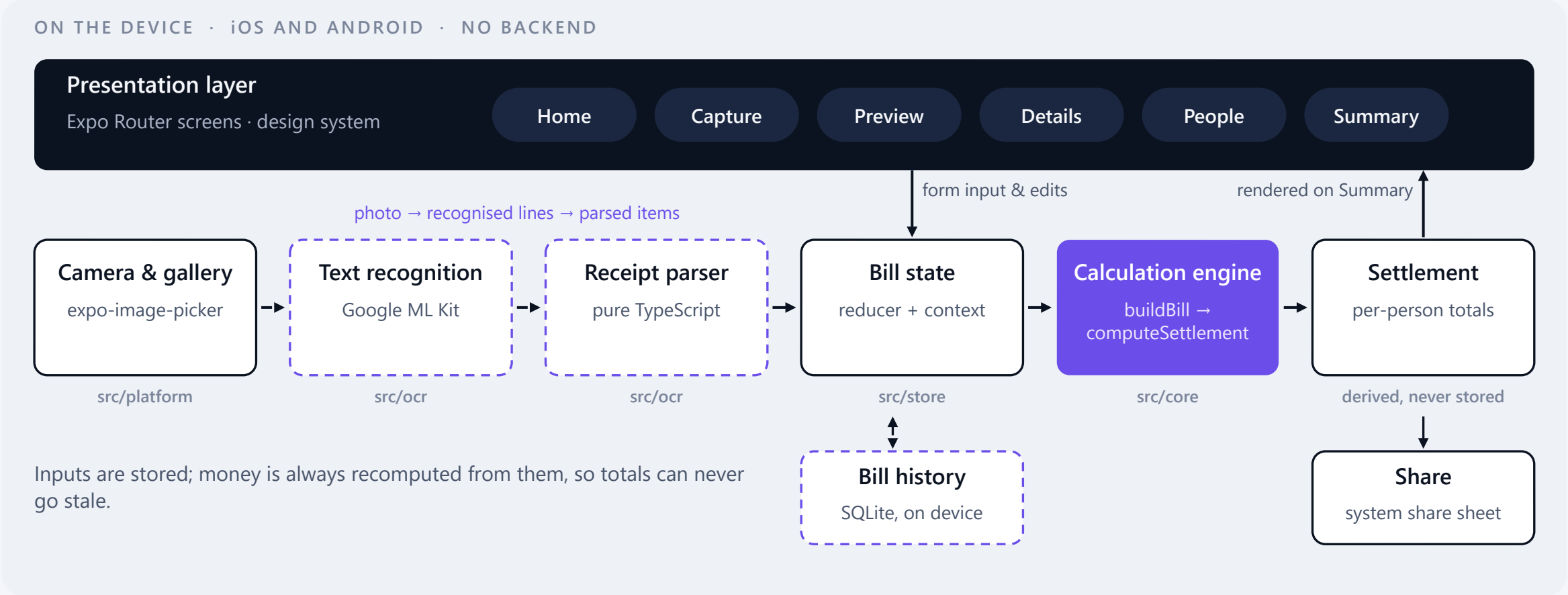
Share summary
Send the split to the group as text

Private by design
On device, no sign-up, nothing uploaded

Implemented Final release

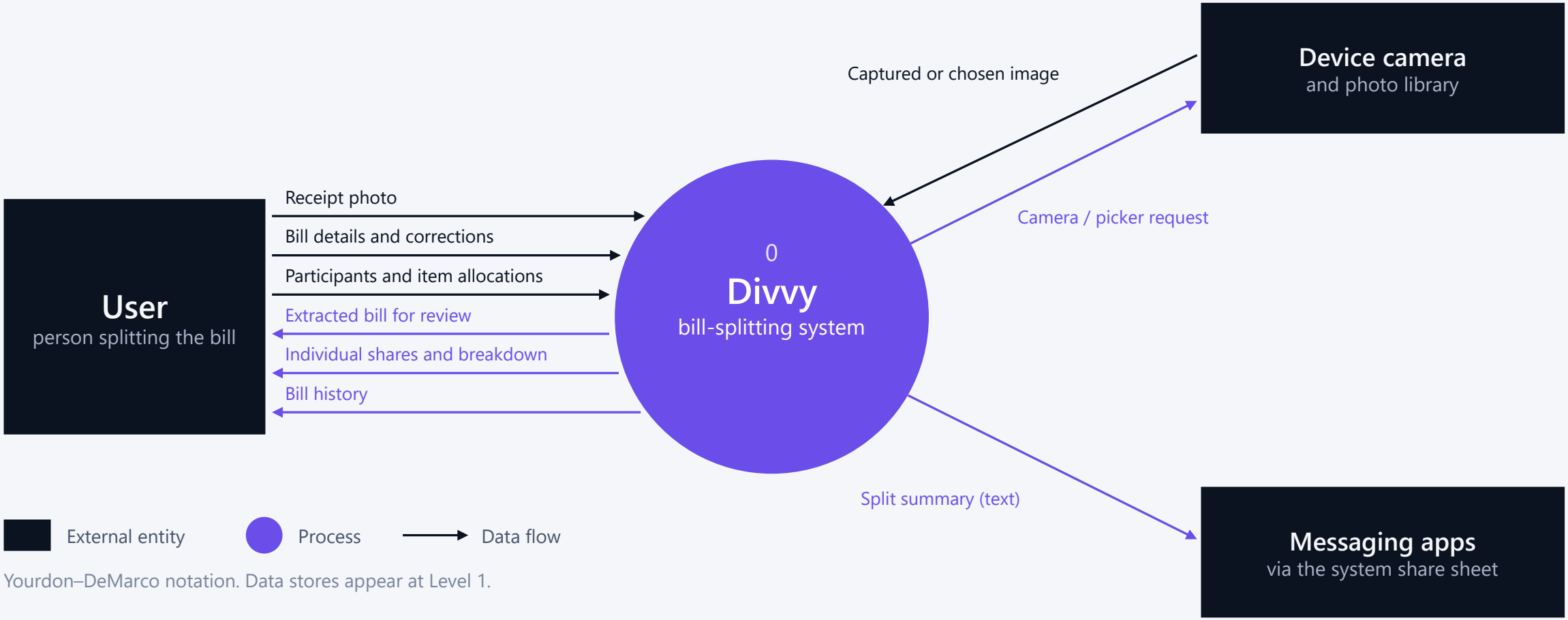
How the system fits together

A layered, fully on-device design. The calculation core is pure TypeScript with no framework imports.



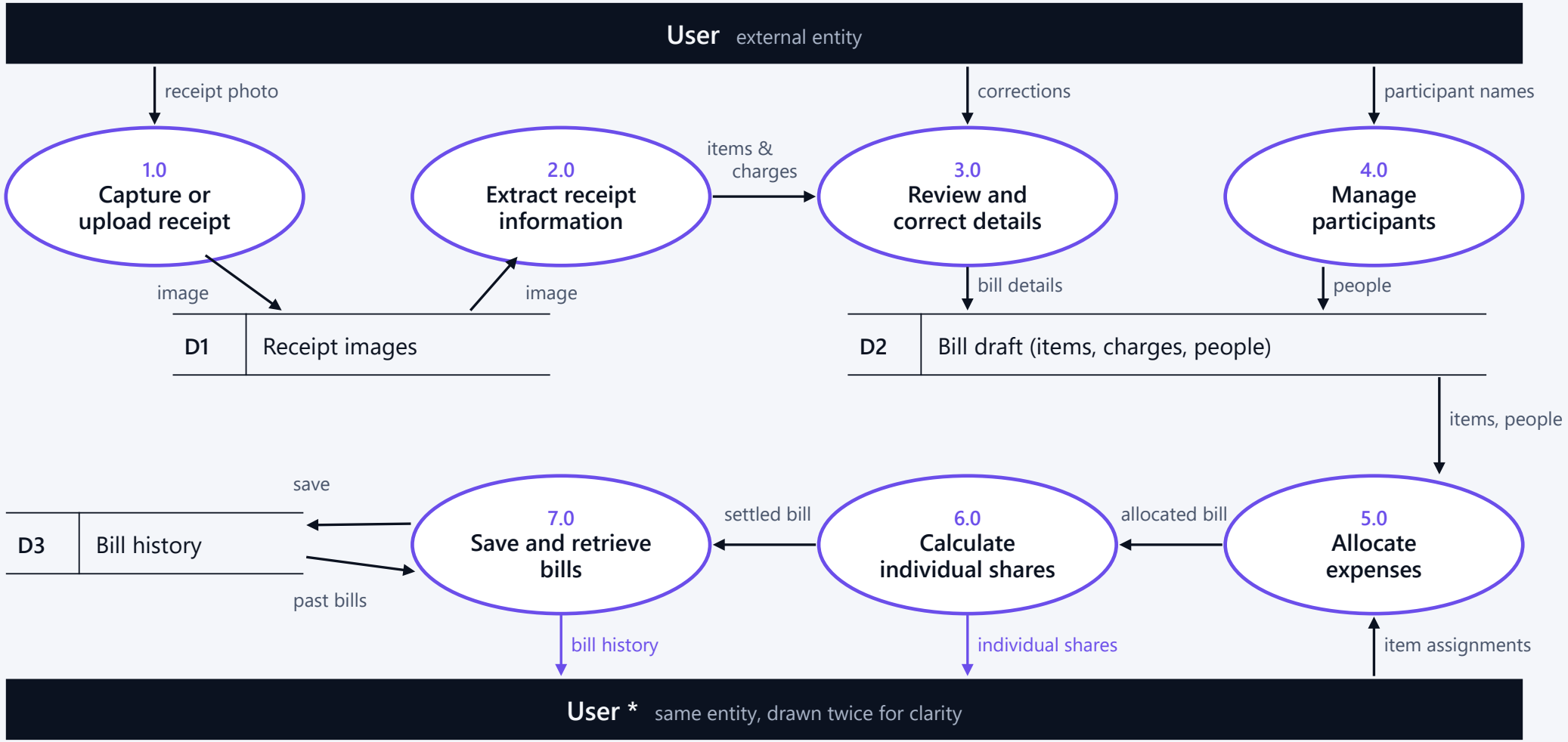
Inputs are stored; money is always recomputed from them, so totals can never go stale.

DFD Level 0 — context diagram



Yourdon–DeMarco notation. Data stores appear at Level 1.

DFD Level 1 — major processes



Implemented Final release

Built on a modern, native-first stack

FRAMEWORK

RN — React Native 0.88 ●
Native UI for iOS and Android from one codebase

EX — Expo SDK 58 ●
Tooling, device APIs and Expo Go on the iPhone

TS — TypeScript ●
Strict types; money is a branded Paise type

APP LAYER

ER — Expo Router ●
File-based, typed navigation between screens

R19 — React 19 ●
UI, with state in a pure reducer + Context

SF — expo-image · expo-symbols ●
Fast images and native SF Symbols icons

DEVICE & DATA

IP — expo-image-picker ●
Camera capture and gallery upload

ML — Google ML Kit ○
On-device text recognition for receipts

SQ — SQLite ○
On-device storage of past bills

QUALITY

JS — Jest + ts-jest ●
274 automated tests, incl. a 1,000-bill property test

ES — ESLint + Prettier ●
Linting and formatting, zero warnings allowed

HB — Hermes ●
JavaScript engine for the mobile bundle

How Divvy calculates each share

1. **Money is an integer**
   Every amount is stored in paise: ₹123.45 is 12345. No floating point ever touches money.

2. **Split each item**
   An item is divided among the people who ordered it. Spare paise go to them in a fixed order.

3. **Apply charges in order**
   Service charge on the items, then GST on items + service, then tip. Percentages round once, half away from zero.

4. **Largest remainder**
   Each charge is shared in proportion to what each person ordered; leftover paise go to the largest remainders.

5. **Reconcile**
   Σ shares = bill total is checked on every bill and proven on 1,000 generated bills.

WORKED EXAMPLE · THE SPICE TABLE

	AA	PR	ZO	Total
Paneer Tikka shared by Aarav, Priya	170.00	170.00	—	**340.00**
Dal Makhani shared by all three	93.34	93.33	93.33	**280.00**
Cold Coffee Zoya	—	—	150.00	**150.00**
Butter Naan ×4 shared by all three	53.34	53.33	53.33	**160.00**
Items subtotal	**316.68**	**316.66**	**296.66**	**930.00**
Service charge 5% of ₹930.00	15.84	15.83	14.83	**46.50**
GST 5% of ₹976.50 = 48.825 → 48.83	16.63	16.63	15.57	**48.83**
Each person pays	**349.15**	**349.12**	**327.06**	**1,025.33**

Σ = 102,533 paise = 93,000 + 4,650 + 4,883 — exactly the bill, as computed by Divvy's engine.

What makes Divvy different

THE GUARANTEE

Σ shares = total

Exactly — not approximately — on every bill. No paisa is created or lost by rounding.

274 automated tests

1,000 generated bills reconciled

8 people per bill

Receipt-first

The bill photo is where every split begins, not an afterthought.

Less typing

Scanning pre-fills items and charges; you confirm instead of typing.

Transparent

Every share is broken down into items, charges and tax.

Exact money

Integer paise and largest-remainder rounding, end to end.

Built for Indian bills

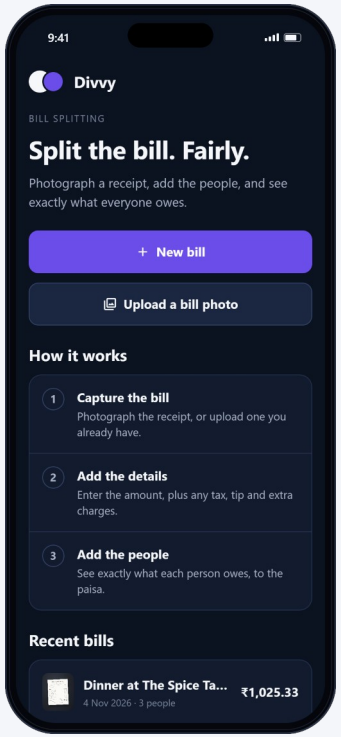
GST on service charge, ₹ with lakh-style grouping.

Private and offline

Runs on the device. No account, no upload, no network.

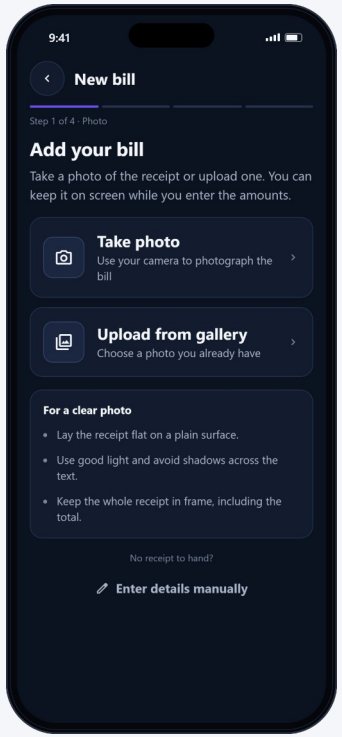
The user journey

1 2 3 4 5 6



Start

Begin a new bill



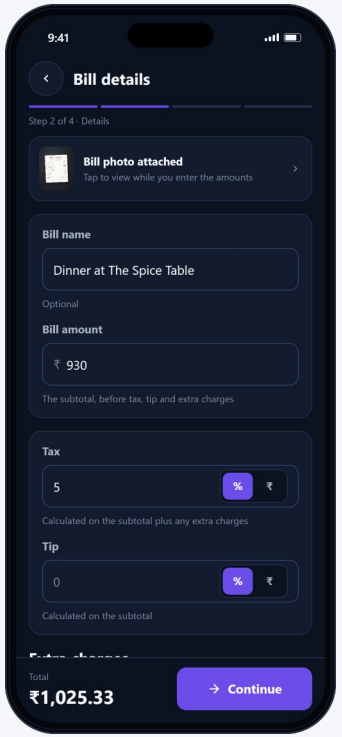
Add the bill

Camera or gallery



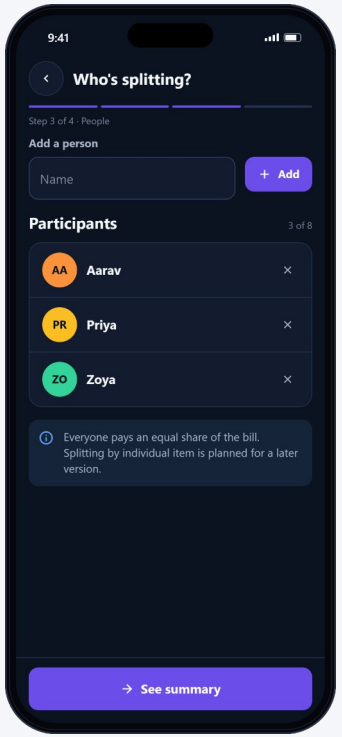
Check the photo

Retake or replace



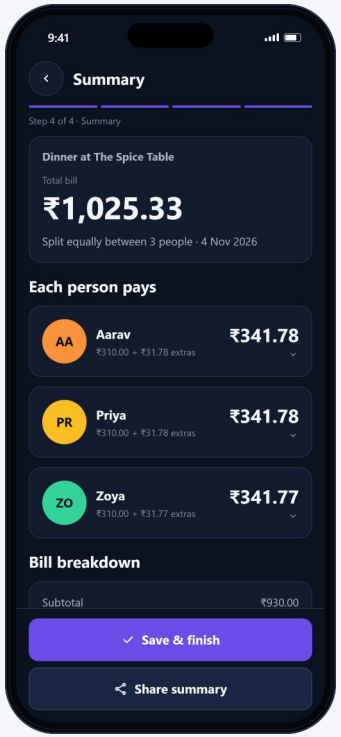
Bill details

Amount, tax, charges



Who's splitting

Add the people



Each person pays

Exact shares

App screenshots of the working flow. Final release adds Scan & review after step 3 and Assign items after step 5.

Where Divvy goes next

Beyond the final release — directions that build on the same exact, on-device core.

Smarter OCR

Tune the parser on a corpus of real Indian receipts; flag low-confidence lines.

Advanced allocation

Drag-to-assign, fractional shares (½, ⅓) and splitting by unit.

Group expenses

Saved groups of friends and flatmates for one-tap setup.

Settlement tracking

Mark who has paid, with a UPI payment link for each person.

History and analytics

Search, filters and monthly spending summaries.

Optional cloud sync

Opt-in backup across devices; on-device stays the default.

Personalisation

Light theme, default tax and tip, saved preferences.

WHAT STAYS THE SAME

Exact money, transparent shares, private by default.

Conclusion

01 Problem

Splitting a shared bill by hand is slow, error-prone and hard to verify.

02 Solution

Divvy turns a receipt photo into an exact, item-by-item split in a few taps.

03 Technical contribution

A pure-TypeScript money engine in integer paise — ordered charges, largest-remainder rounding — proven by 274 tests.

04 Product value

Fair, transparent shares that everyone at the table can trust and check.

Divvy

Thank you

Questions are welcome.

Krish Dwivedi
BSc Computer Science · UIT, Karnavati University
Mobile Application Development · 2026